HIREMATRIX FINAL PROJECT REPORT

GROUP - 6

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Github Link

https://github.com/santhoshkv24/Hire-Matrix

Video Links -

Project Demonstration - *https://drive.google.com/file/d/1ttKAr1XiWusXKxiMaxkWhyXp-95ejnJY/view?usp=drive_link*

Code Demonstration - *https://drive.google.com/file/d/14kB0D9b2Plr2NRQace40E_txYzqlDE3r/view?usp=drive_link*

1. Introduction

HireMatrix is a sophisticated full-stack recruitment management platform designed to centralize and streamline the complete hiring lifecycle for organizations of all sizes. Built with modern web technologies including React 19, Node.js, Express 5, and MongoDB, the system provides a comprehensive solution for managing job postings, candidate applications, interview scheduling, and final hiring decisions.

The platform serves as a critical business tool for HR departments, recruitment teams, and hiring managers by providing a unified interface where all stakeholders can collaborate effectively throughout the hiring process. HireMatrix eliminates the need for multiple disconnected tools like spreadsheets, email threads, and disparate tracking systems by offering a single source of truth for all recruitment activities.

The system's importance extends beyond simple tracking—it introduces intelligent automation through AI-powered resume scoring using Google's Generative AI, automated notification systems, and seamless integration with Google Calendar for interview scheduling. This automation significantly reduces manual effort while improving the quality and consistency of hiring decisions. The platform's role-based access control ensures that each user type (admin, recruiter, hiring manager, interviewer, and applicant) has access to appropriate features and data, maintaining security and workflow integrity throughout the organization.

HireMatrix addresses the common pain points in recruitment: lack of visibility into the hiring pipeline, difficulty in coordinating between multiple stakeholders, inconsistent evaluation criteria, and poor candidate experience. By providing a structured, transparent, and efficient hiring workflow, organizations can make better hiring decisions faster while maintaining a professional image to candidates.

2. Scenario / Use Case Explanation

Scenario 1: Job Posting Creation and Team Assignment

Sarah, a hiring manager at a tech company, needs to fill a Senior Software Engineer position. She logs into HireMatrix and creates a new job posting, specifying the department, location, employment type, required skills, and a detailed description. The system automatically assigns her as the hiring manager for this position. Sarah then uses the hiring teams feature to assign a specific recruiter and additional interviewers to the job. Once configured, the job is marked as "open" and visible to applicants.

Scenario 2: Candidate Application and AI-Powered Screening

John, a software developer, applies for the Senior Software Engineer position through the public jobs page. He registers as an applicant, creates his profile, and uploads his resume. Upon upload, HireMatrix automatically processes the resume using Google's Generative AI to extract relevant information including skills, experience, and qualifications. The AI scoring system evaluates the resume against the job requirements, generating a score from 0-100 along with identified strengths and gaps. This automated screening saves the recruitment team hours of manual review work.

Scenario 3: Application Pipeline Management

The assigned recruiter receives a notification about John's application. They review the AI-generated score and strengths/gaps analysis. Impressed by the candidate's profile, the recruiter moves the application from "applied" to "screening" stage using the intuitive drag-and-drop pipeline interface. After initial screening, the application progresses to the "interview" stage. Throughout this process, the candidate receives automated email notifications about their application status, ensuring a positive candidate experience.

Scenario 4: Interview Scheduling and Feedback Collection

Once an application reaches the interview stage, the hiring manager schedules interviews with multiple team members. HireMatrix integrates with Google Calendar to create meeting events and generate Google Meet links automatically. Each assigned interviewer receives calendar invites with the meeting link and candidate details. After conducting the interview, interviewers submit structured feedback through the platform, rating the candidate on various criteria and providing detailed comments. This centralized feedback collection eliminates email chains and ensures all feedback is properly documented.

Scenario 5: Final Hiring Decision

After all interviews are complete, the hiring manager reviews all feedback and the candidate's complete profile. Using the decision page, they make a final decision: selected, rejected, or pending. If selected, the system can track offer status. Throughout this process, all stakeholders have visibility into the current status, and candidates receive timely updates. The admin can export all hiring data for compliance reporting and analytics.

2. System Requirements

Software Requirements

Backend: - Node.js 18 or higher - npm 9 or higher - MongoDB 6.0+ (local or cloud-hosted) - Environment variables configuration (.env file)

Frontend: - Modern web browser (Chrome, Firefox, Safari, Edge) - Vite development server for development - React 19 runtime

Third-Party Services: - Google Generative AI API (for resume scoring) - Google Calendar API (for interview scheduling) - Google Meet API (for meeting link generation) - Nodemailer with SMTP configuration (for email notifications)

Hardware Requirements

Development Environment: - CPU: Dual-core processor (2.0 GHz or higher recommended) - RAM: Minimum 4GB (8GB recommended) - Storage: 500MB for application + space for uploaded resumes - Network: Stable internet connection for external API calls

Production Environment: - CPU: 4 cores or higher recommended - RAM: 8GB minimum, 16GB recommended - Storage: SSD recommended for database performance - Network: High-speed internet connection - Additional storage for uploaded resume files (scales with usage)

Database: - MongoDB storage: Minimum 1GB for application data - File storage: Approximately 1-2MB per uploaded resume - Recommended: MongoDB Atlas or equivalent cloud hosting for production

3. System Architecture

HireMatrix follows a modern three-tier architecture pattern with clear separation of concerns between presentation, application, and data layers. The system is built using a monolithic repository (monorepo) structure containing both backend and frontend applications that communicate through a RESTful API.

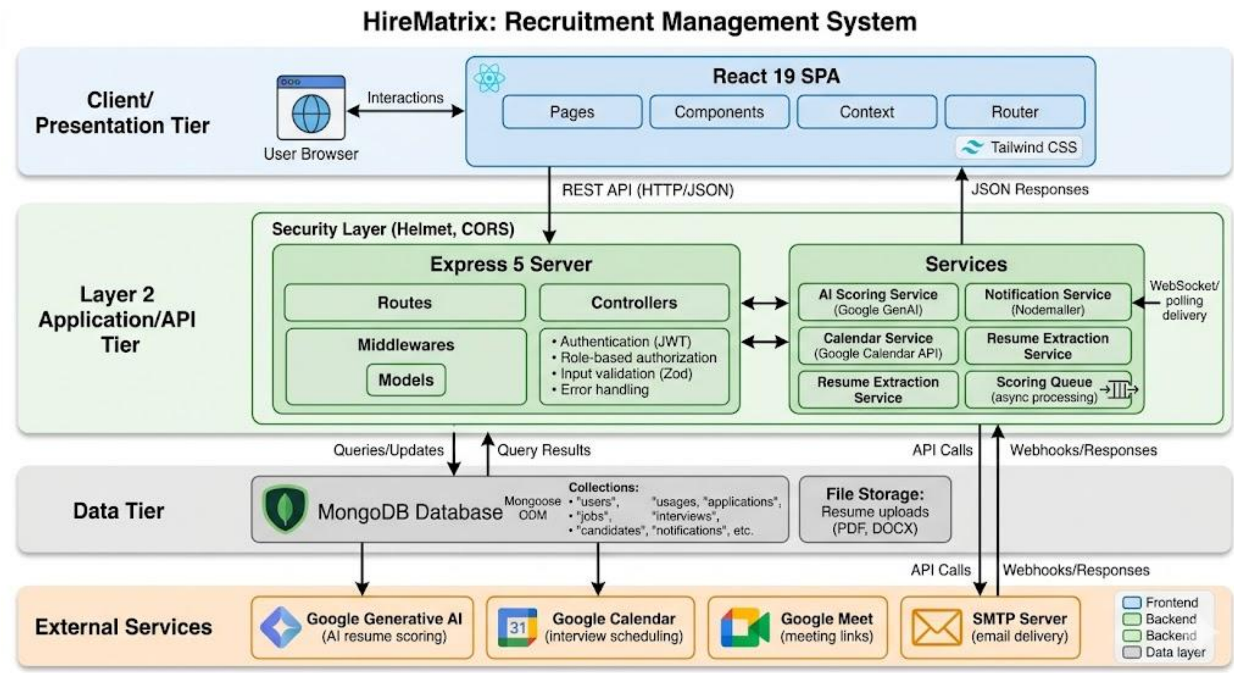
The frontend layer, built with React 19 and Vite, serves as the presentation tier. It handles all user interactions, form validations, and UI rendering. The application uses React Router for client-side routing and implements role-based navigation where different user types see different interfaces. State management is handled through React Context API, particularly for authentication state. The frontend communicates with the backend through HTTP requests using the Fetch API, with all API calls centralized in utility functions for maintainability.

The backend layer, built with Express 5 and Node.js, forms the application tier. It follows a modified MVC (Model-View-Controller) pattern where controllers handle business logic, models define data schemas and relationships, and routes define API endpoints. The backend implements middleware for cross-cutting concerns: CORS for security, Helmet for HTTP security headers, Morgan for request logging, and custom middleware for authentication and role-based authorization. All requests pass through this middleware chain before reaching controllers.

The data tier consists of MongoDB with Mongoose ODM for object-document mapping. MongoDB was chosen for its flexibility in handling varying document structures, which is ideal for candidate profiles and application data that may have varying attributes. Mongoose schemas define the structure and relationships between entities, with proper indexing for query performance. The database layer also includes file storage for uploaded resumes in the local filesystem (configurable for cloud storage).

External integrations form an important part of the architecture. The Google Generative AI service processes resume text to extract skills and generate scores. Google Calendar API creates and manages interview events. Google Meet integration generates virtual meeting links. The notification system uses Nodemailer for email delivery and maintains an in-app notification system for real-time updates.

The application pipeline uses a queue-based architecture for AI scoring, allowing asynchronous processing of resume evaluations without blocking user requests. This ensures the system remains responsive even during peak usage periods.



5. ER Diagram

The HireMatrix database schema consists of eleven core entities that work together to manage the complete hiring lifecycle.

User is the central authentication entity, storing name, email, password hash, and role assignments. Users can have multiple roles and maintain a last login timestamp for security auditing.

Role defines the permission structure with keys like admin, recruiter, hiring_manager, interviewer, and applicant. Each role has associated permissions that control access to features.

Job represents open positions with fields for title, department, location, employment type, description, required skills, and status (draft, open, closed, cancelled). Jobs are created by users and can have assigned recruiters and hiring managers.

Candidate stores applicant-specific information including phone, years of experience, skills array, source, and notes. Each candidate is linked to a User account and has a reference to their latest resume.

Resume manages uploaded resume files with metadata including original name, MIME type, file size, storage path, and extracted text. The schema tracks upload status from uploaded through processing to processed or failed.

Application is the central tracking entity connecting candidates to jobs. It maintains the current pipeline stage (applied, screening, interview, offer, hired, rejected), AI scoring data, and final decision status. Applications can be archived with proper audit trails.

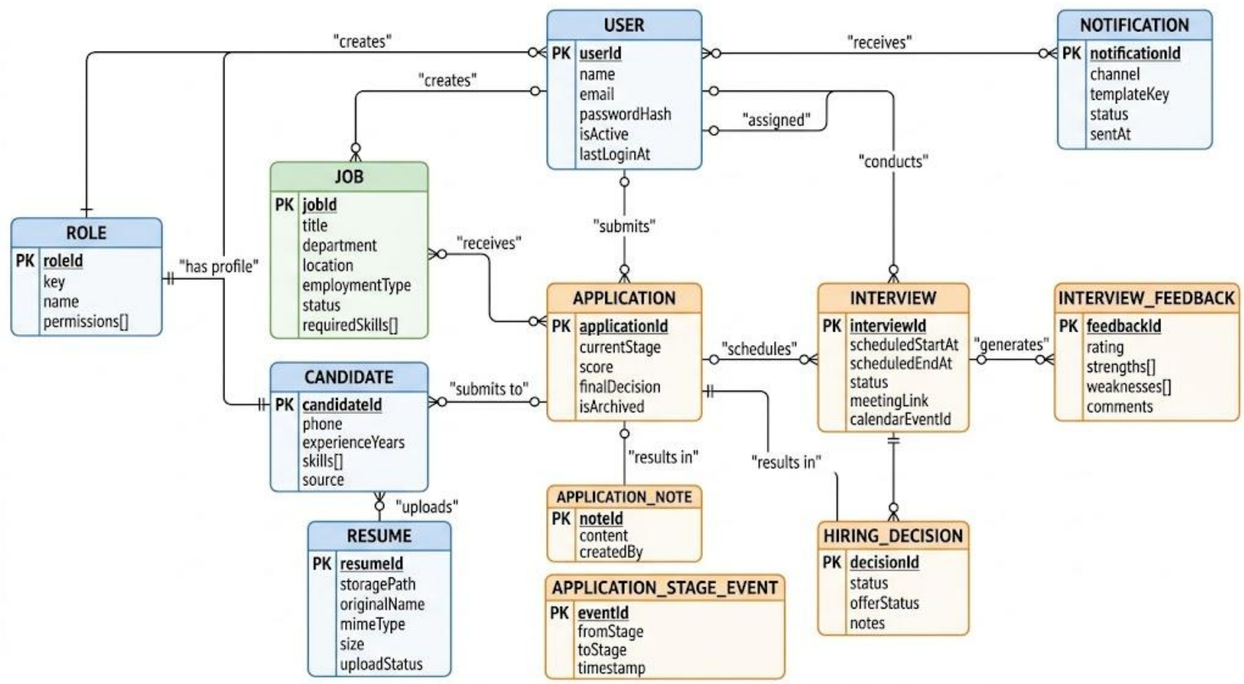
Interview schedules interview sessions linking applications, candidates, and interviewers. It stores scheduled times, timezone, status, meeting details (Google Meet or manual), calendar event IDs, and cancellation information.

InterviewFeedback captures structured feedback from interviewers including ratings, strengths, weaknesses, and recommendations.

HiringDecision records the final hiring decision with status, offer details, and notes.

Notification manages both email and in-app notifications with template support, delivery status tracking, and retry logic.

ApplicationNote, **ApplicationStageEvent**, and **ScoringJob** provide supporting functionality for notes, stage change tracking, and AI scoring job management.



6. Key Features

Role-Based Access Control (RBAC)

HireMatrix implements comprehensive role-based access control with five distinct user types. Admins have full system access including user management, job creation, candidate management, and system configuration. Recruiters manage the day-to-day recruitment process including reviewing applications, moving candidates through the pipeline, and coordinating interviews. Hiring Managers oversee specific positions, make final hiring decisions, and collaborate with recruiters. Interviewers participate in the interview process by providing feedback on candidates. Applicants have access to view job postings, submit applications, upload resumes, and track their application status. Each role sees a customized interface with appropriate navigation and functionality.

Job Management

The system provides complete job posting lifecycle management. Users can create detailed job postings with title, department, location, employment type (full-time, part-time, contract, internship), description, and required skills. Jobs can be in draft, open, closed, or cancelled status. The hiring teams feature allows assignment of specific recruiters and hiring managers to each position. Jobs can be made publicly visible for applicant submissions or kept internal for reference.

Candidate Management

HireMatrix maintains comprehensive candidate profiles including personal information, contact details, years of experience, skills inventory, and application history. The system automatically extracts information from uploaded resumes using AI, populating candidate profiles with minimal manual entry. Candidates can submit multiple applications across different positions while maintaining a single profile. The system tracks all candidate interactions and application history.

AI-Powered Resume Scoring

One of HireMatrix's most innovative features is its AI-powered resume scoring system. When a candidate uploads a resume, the system uses Google Generative AI to analyze the document. The AI extracts skills, experience details, and qualifications, then generates a numerical score (0-100) based on alignment with job requirements. The system also identifies specific strengths and gaps, providing actionable insights for recruiters. This automated scoring happens asynchronously through a queue system, ensuring the main application remains responsive.

Application Pipeline Management

Applications move through a structured pipeline with stages: applied, screening, interview, offer, hired, and rejected. The pipeline stage is visually represented and can be updated through drag-and-drop functionality. Each stage transition is tracked with timestamps for analytics. The system enforces valid stage transitions to maintain data integrity. Recruiters can view all applications in a Kanban-style board or list view, filter by various criteria, and bulk-update stages.

Interview Scheduling and Management

The interview management system integrates with Google Calendar to schedule interviews at mutual convenient times. When scheduling, the system automatically creates calendar events, generates Google Meet links (or allows manual meeting links), and sends invitations to all participants. Interviewers receive notifications with candidate details and meeting links. The system tracks interview status (scheduled, rescheduled, cancelled, completed) and maintains a complete history.

Structured Interview Feedback

After interviews, interviewers submit structured feedback through the platform. The feedback form includes rating scales, strengths assessment, weakness identification, and free-form comments. This standardized approach ensures consistent evaluation across all candidates and interviewers. All feedback is centralized and easily accessible to hiring managers during decision-making.

Final Decision Tracking

The decision module allows hiring managers to make final hiring decisions with three possible statuses: pending, selected, or rejected. For selected candidates, the system tracks offer status and includes notes for documentation. All decisions are timestamped and attributed to the deciding user for audit purposes.

Notification System

HireMatrix maintains both in-app and email notifications. The notification system uses templates for consistency and supports various events: application received, stage changes, interview invitations, interview reminders, and decision notifications. The system tracks notification delivery status with retry logic for failed deliveries. Users can view their notification history and mark notifications as read.

Data Export

The export functionality allows administrators to export hiring data in structured formats for reporting, compliance, and analytics. Exports can include job postings, candidate information, application history, interview feedback, and hiring decisions.

Public Job Board

Organizations can share a public-facing job board where candidates can view open positions and submit applications. The public interface requires registration and provides candidates with application status tracking.

7. Roles and Responsibilities

Admin - Create and manage user accounts for all roles - Configure system settings and permissions - Create and manage job postings - Assign recruiters and hiring managers to positions - Access all applications and candidate data - Export hiring data and generate reports - Manage hiring teams and organizational structure - View system-wide analytics and metrics

Recruiter - Create and manage job postings - Review incoming applications - Move candidates through the pipeline stages - Schedule interviews with candidates - Coordinate with hiring managers and interviewers - Communicate with candidates throughout the process - Access candidate profiles and resumes - Submit feedback and notes on candidates

Hiring Manager - Create job postings for open positions - Review applications for assigned positions - Participate in interviews - Make final hiring decisions (select/reject) - Track offer status and negotiate terms - Collaborate with recruiters on candidate evaluation - View hiring analytics for their department

Interviewer - View scheduled interviews with candidate details - Access candidate resumes and profiles before interviews - Submit structured feedback after interviews - View interview calendar and availability - Communicate with hiring team members

Applicant - View public job postings - Register and create profile - Upload and manage resume - Submit applications for open positions - Track application status - Receive notifications about application updates - Participate in interview scheduling

8. User Flow

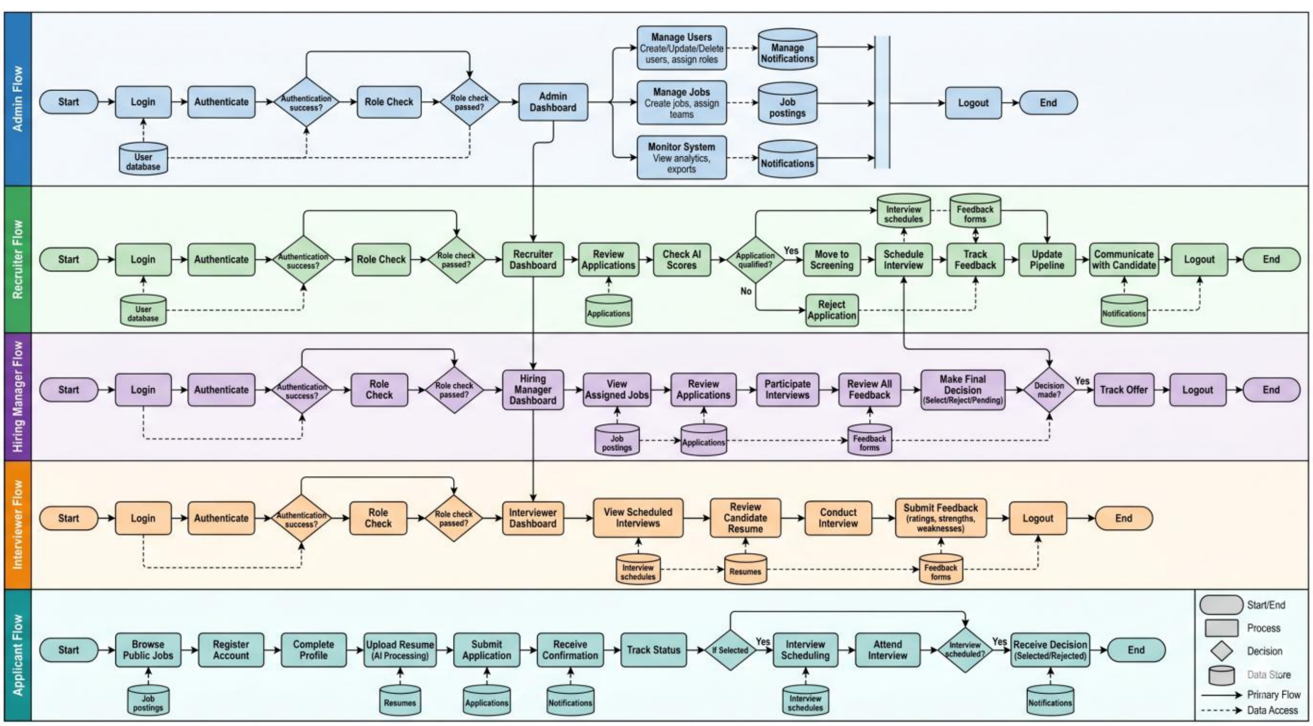
Admin User Flow: 1. Admin logs into the system 2. Views dashboard with hiring metrics 3. Creates new user accounts or manages existing users 4. Creates new job postings or manages existing jobs 5. Assigns recruiters and hiring managers to positions 6. Monitors hiring pipeline across all positions 7. Exports hiring data for reporting

Recruiter User Flow: 1. Recruiter logs into the system 2. Views assigned jobs and applications dashboard 3. Reviews new applications in the pipeline 4. Evaluates AI-generated resume scores and insights 5. Moves qualified candidates to screening stage 6. Schedules interviews for selected candidates 7. Coordinates with interviewers and hiring managers 8. Tracks applications through to final decision

Hiring Manager User Flow: 1. Hiring manager logs into the system 2. Views jobs they're managing 3. Reviews applications for their open positions 4. Participates in interview process 5. Reviews all interview feedback 6. Makes final hiring decision 7. Tracks offer status and onboarding

Interviewer User Flow: 1. Interviewer receives interview invitation notification 2. Logs into system to view interview details 3. Reviews candidate resume and profile 4. Conducts interview (via Google Meet or in-person) 5. Submits structured feedback through the platform 6. Views upcoming interview schedule

Applicant User Flow: 1. Candidate visits public job board 2. Browses available positions 3. Registers for an account 4. Completes profile with personal information 5. Uploads resume (automatically parsed by AI) 6. Submits application for desired position 7. Receives confirmation and status updates 8. Participates in interview process if selected 9. Receives final decision notification



9. Architecture Pattern

HireMatrix implements a variation of the Model-View-Controller (MVC) architectural pattern, adapted for a modern single-page application (SPA) frontend with a REST API backend.

Backend Architecture (Express.js):

The backend follows a clear MVC-inspired structure:

Routes Layer: Defines all API endpoints organized by resource type (auth, jobs, candidates, applications, interviews, etc.). Each route file maps HTTP verbs and paths to controller functions.

Controllers Layer: Contains business logic for each domain. Controllers handle request validation, interact with models for data operations, and return appropriate responses. Each controller is organized by domain (authController, jobController, candidateController, applicationController, interviewController, etc.).

Models Layer: Mongoose schemas and models define data structure, relationships, and validation rules. Models encapsulate database operations and business logic related to data persistence.

Middlewares: Cross-cutting concerns are handled by middleware functions including authentication (JWT verification), authorization (role-based access control), error handling, input validation (using Zod), and logging.

Services Layer: Additional business logic that doesn't fit neatly into controllers, including AI scoring service, notification service, calendar service, and bootstrap service for system initialization.

Frontend Architecture (React):

The frontend uses a component-based architecture with React:

Pages: Top-level components corresponding to routes (JobsPage, CandidatesPage, ApplicationsPage, etc.). Each page component handles data fetching and layout for a specific view.

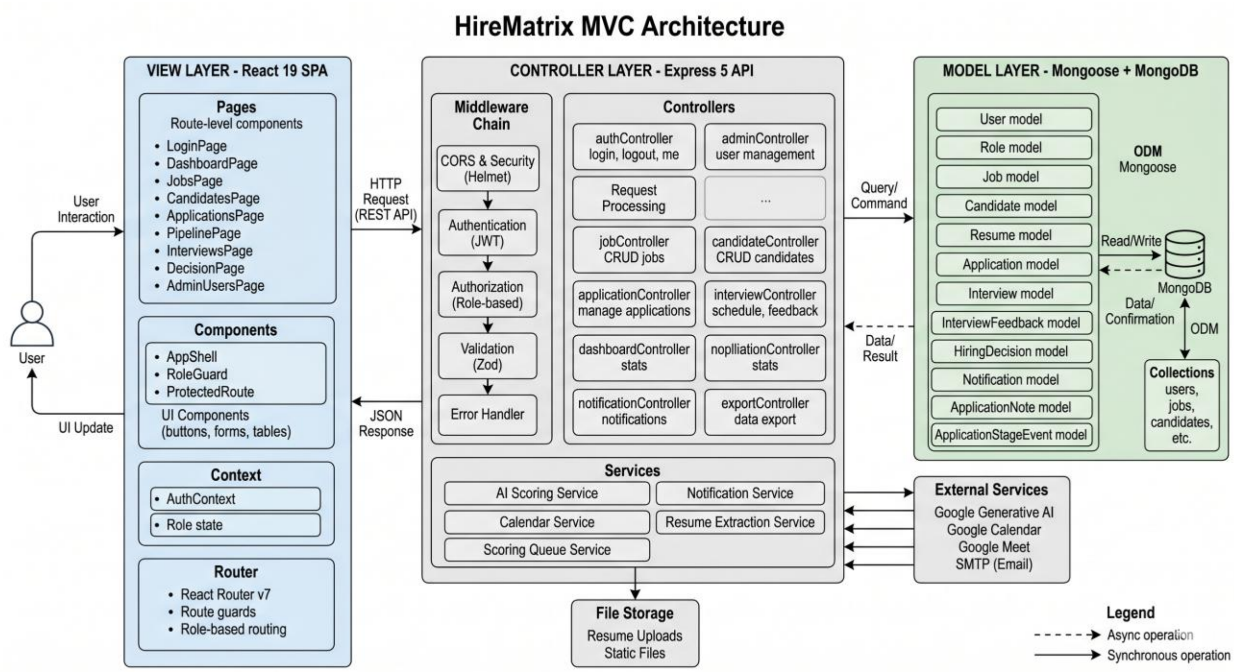
Components: Reusable UI components (AppShell, navigation, forms, tables, modals, etc.). Components are organized in the components directory with UI components in a subdirectory.

Context: React Context API for global state management, particularly authentication state and user information.

Routing: React Router handles client-side routing with protected routes that check authentication and role-based access.

Utilities: Helper functions for API calls, formatting, routing logic, and other shared functionality.

This architecture provides clear separation of concerns, maintainability, and scalability. The frontend and backend communicate through a well-defined REST API contract, allowing independent development and deployment.



10. Project Setup

Prerequisites

Before setting up HireMatrix, ensure you have the following installed: - Node.js version 18 or higher - npm version 9 or higher - MongoDB instance (local or cloud-hosted) - Git for version control

Backend Setup

1. Navigate to the backend directory:
   cd backend
2. Install dependencies:
   npm install
3. Create a *.env* file in the backend directory with the following required variables:
   PORT=4000
   MONGO_URI=mongodb://127.0.0.1:27017/hirematrix
   CORS_ORIGIN=http://localhost:5173
   JWT_SECRET=your-super-secret-jwt-key
   JWT_REFRESH_SECRET=your-refresh-secret-key
   ADMIN_EMAIL=admin@hirematrix.local
   ADMIN_PASSWORD=ChangeMe123!
   APP_FRONTEND_URL=http://localhost:5173
4. Start the development server:
   npm run dev

The backend will start on http://localhost:4000

Frontend Setup

1. Navigate to the frontend directory:
   cd frontend
2. Install dependencies:
   npm install
3. Create a *.env* file (optional):
   VITE_API_URL=http://localhost:4000/api/v1
4. Start the development server:
   npm run dev

The frontend will start on http://localhost:5173

Production Build

1. Build the frontend:
   cd frontend
   npm run build
2. Start the backend in production mode:

```
cd backend
npm start
```

11. Backend Development

Directory Structure

```
backend/
├── src/
│   ├── config/
│   │   ├── constants.js    # System constants and enums
│   │   ├── database.js     # MongoDB connection
│   │   ├── env.js          # Environment variable validation
│   │   └── mailer.js       # Email configuration
│   ├── controllers/        # Business logic handlers
│   ├── middlewares/        # Express middlewares
│   ├── models/             # Mongoose schemas
│   ├── routes/             # API route definitions
│   ├── services/           # Business logic services
│   ├── app.js              # Express app setup
│   └── server.js           # Server entry point
├── uploads/                # File storage
└── package.json
```

API Endpoints

Authentication: - POST /api/v1/auth/login - User login - POST /api/v1/auth/logout - User logout - GET /api/v1/auth/me - Get current user

Admin: - GET /api/v1/admin/users - List all users - POST /api/v1/admin/users - Create user - PUT /api/v1/admin/users/:id - Update user - DELETE /api/v1/admin/users/:id - Delete user

Jobs: - GET /api/v1/jobs - List jobs - POST /api/v1/jobs - Create job - GET /api/v1/jobs/:id - Get job details - PUT /api/v1/jobs/:id - Update job - DELETE /api/v1/jobs/:id - Delete job

Candidates: - GET /api/v1/candidates - List candidates - POST /api/v1/candidates - Create candidate - GET /api/v1/candidates/:id - Get candidate - PUT /api/v1/candidates/:id - Update candidate - POST /api/v1/candidates/:id/resume - Upload resume

Applications: - GET /api/v1/applications - List applications - POST /api/v1/applications - Create application - GET /api/v1/applications/:id - Get application - PUT /api/v1/applications/:id/stage - Update stage - DELETE /api/v1/applications/:id - Delete application

Interviews: - GET /api/v1/interviews - List interviews - POST /api/v1/interviews - Schedule interview - PUT /api/v1/interviews/:id - Update interview - POST /api/v1/interviews/:id/feedback - Submit feedback

Dashboard: - GET /api/v1/dashboard/stats - Get dashboard statistics

Notifications: - GET /api/v1/notifications - List notifications - PUT /api/v1/notifications/:id/read - Mark as read

Exports: - GET /api/v1/exports/jobs - Export jobs data - GET /api/v1/exports/candidates - Export candidates data

Authentication & Authorization

HireMatrix uses JWT (JSON Web Tokens) for authentication. When a user logs in, the system generates an access token and refresh token. The access token is included in the Authorization header of subsequent requests.

Role-based authorization is implemented through middleware that checks the user's assigned roles against required roles for each endpoint.

Services

BootstrapService: Initializes system defaults including roles and admin user on first run.

ScoringService: Handles AI-powered resume scoring using Google Generative AI.

ScoringQueueService: Manages asynchronous queue for processing resume scoring jobs.

NotificationService: Handles sending of email and in-app notifications.

CalendarService: Integrates with Google Calendar for interview scheduling.

ResumeExtractionService: Extracts text and data from uploaded resume files (PDF, DOCX).

12. Database Design

Core Collections

users: Stores user authentication and profile information - Fields: name, email, passwordHash, roles (array), isActive, lastLoginAt - Indexes: email (unique), isActive

roles: Defines system roles and permissions - Fields: key, name, permissions (array) - Indexes: key (unique)

jobs: Job postings and descriptions - Fields: title, department, location, employmentType, description, requiredSkills, status, createdBy, assignedRecruiter, assignedHiringManager, targetStartDate - Indexes: status, department + status compound

candidates: Candidate profiles - Fields: userId, phone, experienceYears, skills (array), source, notes, latestResumeId, createdBy - Indexes: userId (unique), skills (text)

resumes: Uploaded resume files - Fields: candidateId, storagePath, originalName, mimeType, size, extractedText, extractionError, uploadStatus, uploadedBy - Indexes: candidateId + createdAt

applications: Application tracking - Fields: jobId, candidateId, submittedByUserId, currentStage, stageChangedAt, assignedRecruiter, assignedHiringManager, score (object), finalDecision (object), isArchived - Indexes: jobId, candidateId, currentStage, jobId + currentStage compound, candidateId + jobId + isArchived (unique partial)

interviews: Interview scheduling - Fields: applicationId, candidateId, interviewerId, scheduledStartAt, scheduledEndAt, timezone, status, notes, meetingProvider, meetingLink, calendarEventId, attendees, cancelledReason, createdBy - Indexes: applicationId, scheduledStartAt, interviewerId + scheduledStartAt

notifications: Notification queue and history - Fields: recipientUserId, recipientEmail, channel, templateKey, payload, status, retries, lastError, sentAt, readAt - Indexes: status, recipientEmail + createdAt, recipientUserId + readAt + createdAt

Relationships

- User belongs to many Roles (through roles array)
- Job belongs to User (createdBy)
- Candidate belongs to User (userId)
- Resume belongs to Candidate (candidateId)
- Application belongs to Job (jobId) and Candidate (candidateId)
- Interview belongs to Application (applicationId), Candidate (candidateId), and User (interviewerId)
- Notification belongs to User (recipientUserId)

Data Integrity

The application uses Mongoose's schema validation, required fields, and enum constraints to ensure data integrity. Compound indexes optimize common query patterns. Partial indexes on applications ensure unique active applications per candidate-job combination while allowing historical duplicates when archived.

13. Frontend Development

Directory Structure

```
frontend/
├── src/
│   ├── components/
│   │   ├── ui/          # Base UI components
│   │   ├── AppShell.jsx    # Main layout shell
│   │   ├── ApplicantShell.jsx
│   │   ├── ProtectedRoute.jsx
│   │   └── RoleGuard.jsx
│   ├── pages/          # Page components
│   ├── context/         # React Context providers
│   ├── utils/          # Helper functions
│   ├── lib/            # Library configurations
│   └── App.jsx          # Root component with routing
```

Key Pages

LoginPage: User authentication with email/password form

DashboardPage: Overview of hiring metrics and activity

JobsPage: Job listing, creation, and management

CandidatesPage: Candidate database and profile management

ApplicationsPage: Application tracking and pipeline view

PipelinePage: Kanban-style pipeline with drag-and-drop stage management

InterviewsPage: Interview scheduling and management

InterviewFeedbackPage: Feedback submission form

DecisionPage: Final hiring decision interface

AdminUsersPage: User management (admin only)

ExportsPage: Data export functionality

NotificationsPage: Notification center

State Management

The application uses React Context for global state: - AuthContext: User authentication state and methods - Role-based rendering through RoleGuard component

Routing

React Router v7 handles navigation with: - Protected routes requiring authentication - Role guards for role-specific pages - Dynamic routing based on user roles - Default landing pages per role

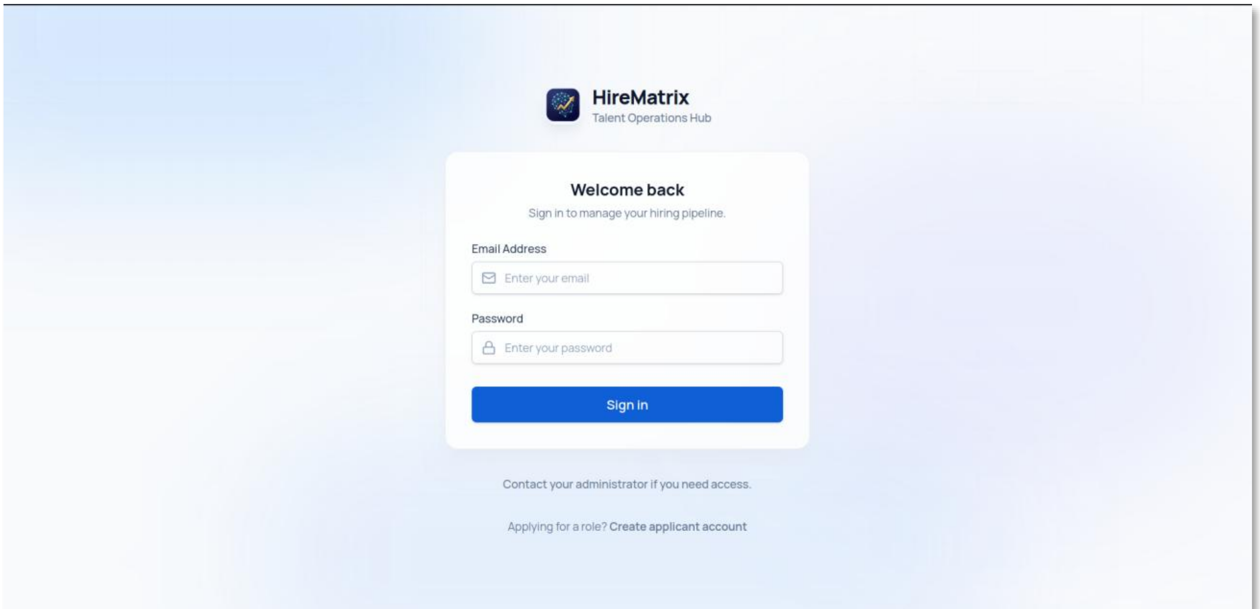
Styling

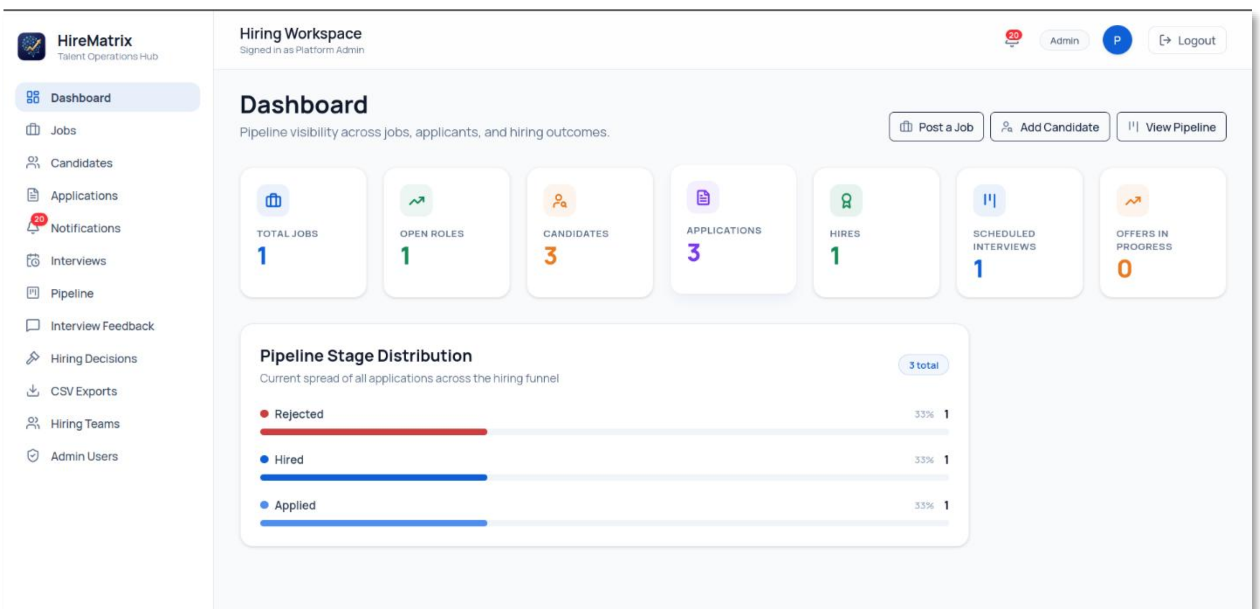
The application uses: - Tailwind CSS v4 for utility-first styling - Custom CSS for specific component styles - Responsive design for mobile compatibility - Lucide React icons throughout

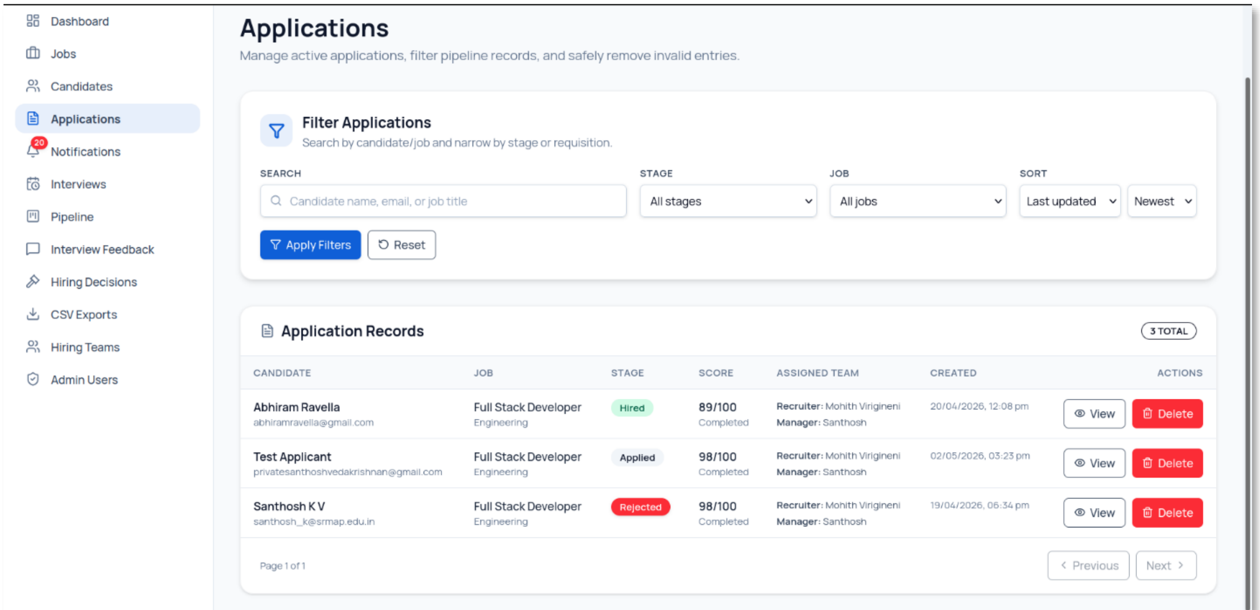
Key Dependencies

- React 19: UI framework
- React Router DOM 7: Routing
- Tailwind CSS 4: Styling
- Framer Motion: Animations
- Dnd-kit: Drag and drop functionality
- Lucide React: Icon library
- Class Variance Authority: Component variant management

14. Output Screenshots






Dashboard
Jobs
Candidates
Applications
20
Notifications
Interviews
Pipeline
Interview Feedback
Hiring Decisions
CSV Exports
Hiring Teams
Admin Users
Applications
Manage active applications, filter pipeline records, and safely remove invalid entries.
Filter Applications
Search by candidate/job and narrow by stage or requisition.
SEARCH
Candidate name, email, or job title
STAGE
All stages
JOB
All jobs
SORT
Last updated
Newest
Apply Filters
Reset
Application Records
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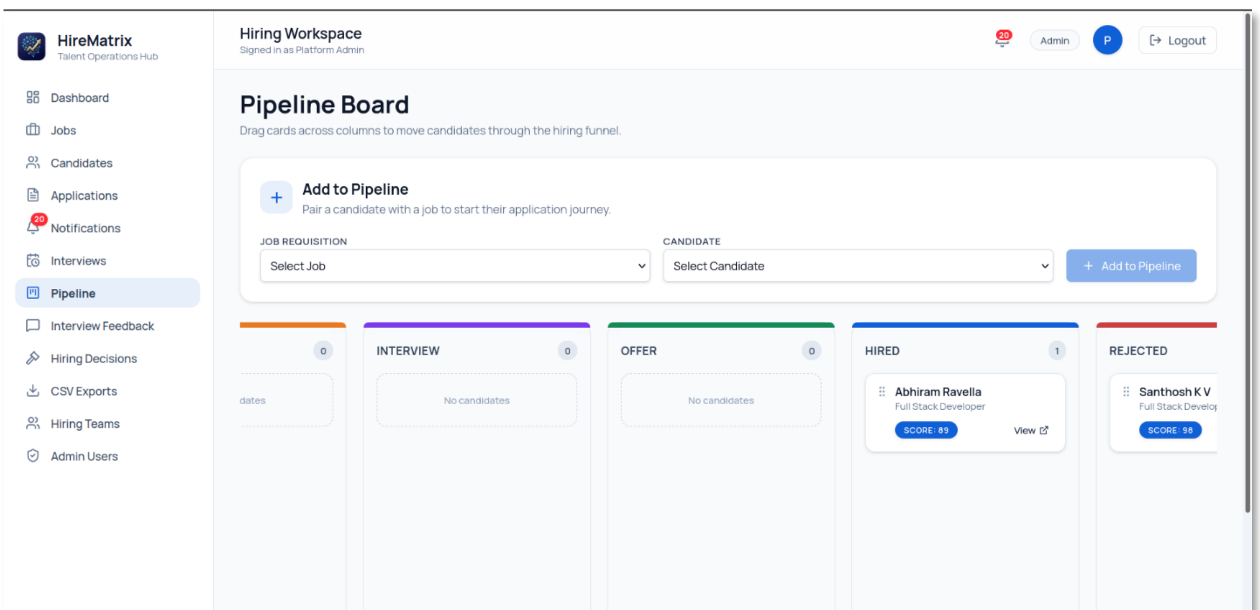
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Hiring Workspace
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Dashboard
Jobs
Candidates
Applications
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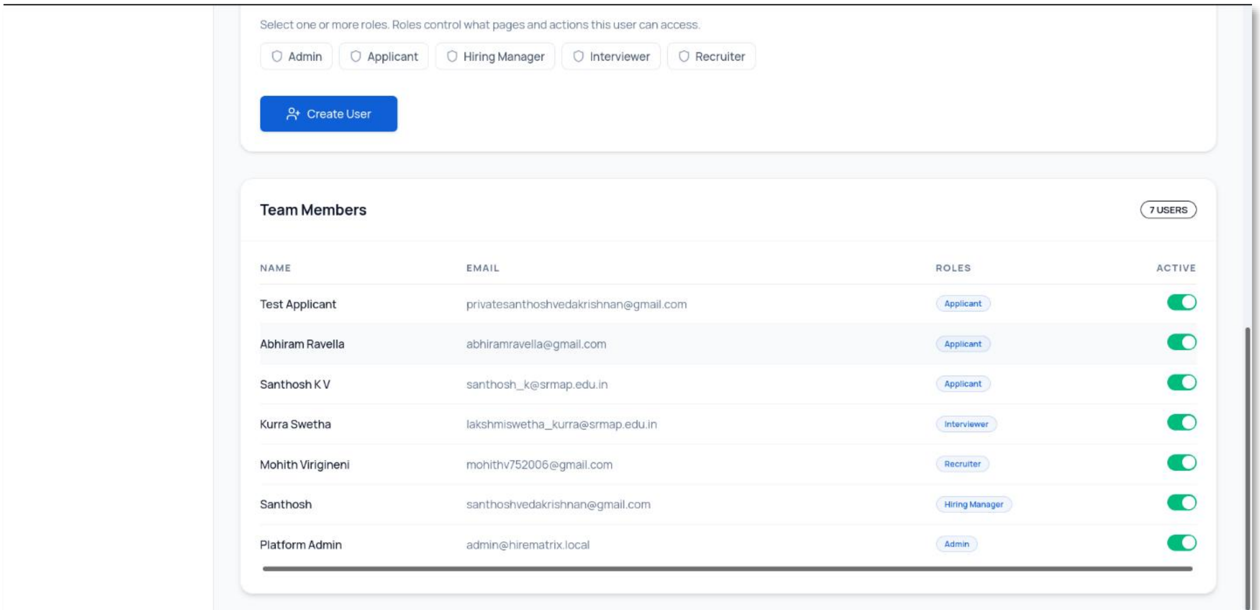
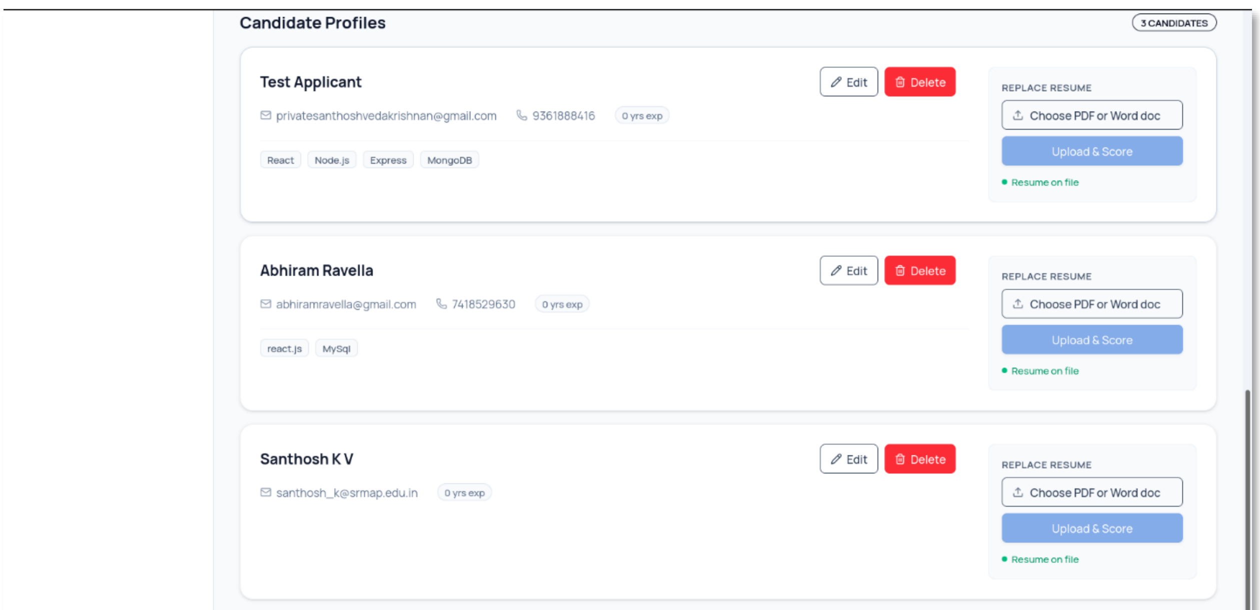
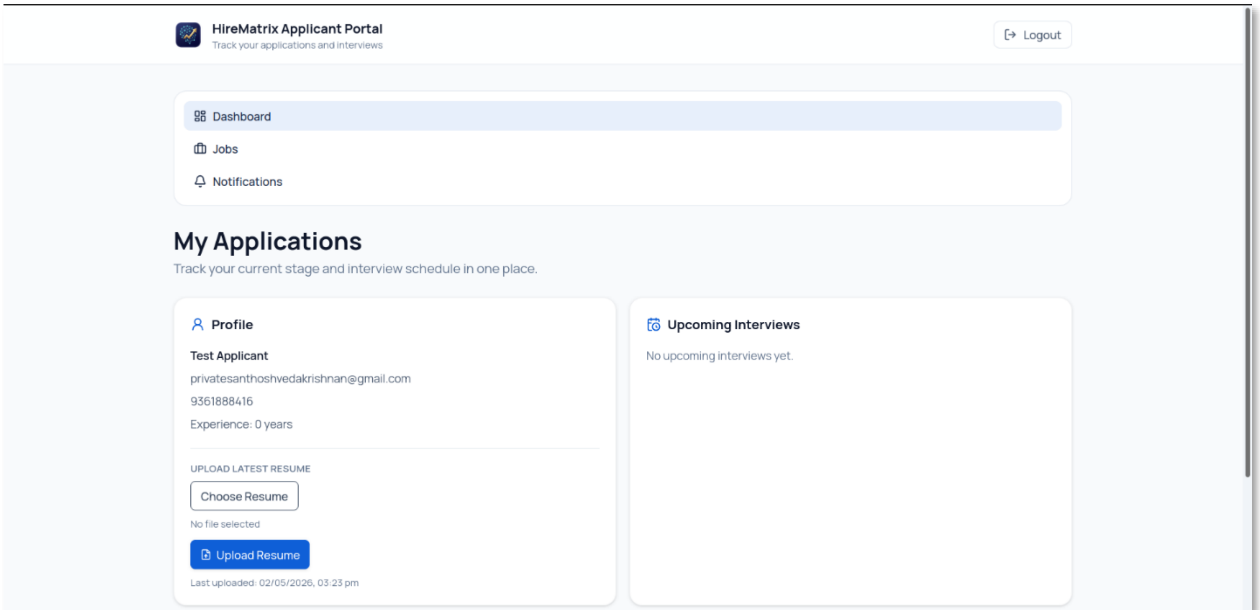
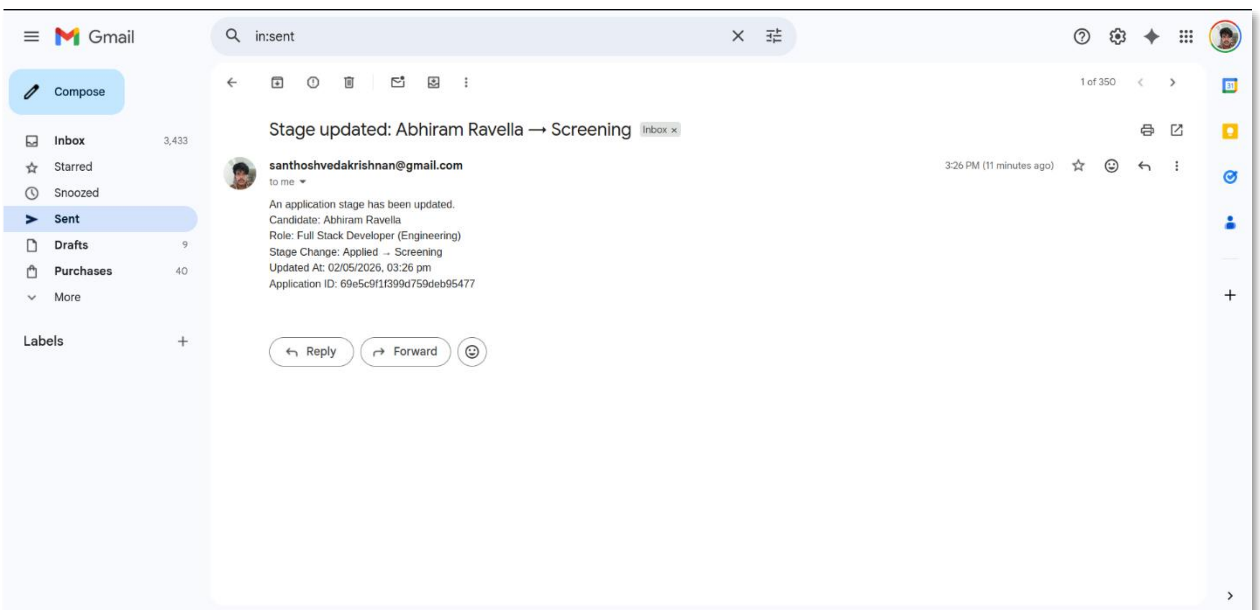
Notifications
Interviews
Pipeline
Interview Feedback
Hiring Decisions
CSV Exports
Hiring Teams
Admin Users
Pipeline Board
Drag cards across columns to move candidates through the hiring funnel.
Add to Pipeline
Pair a candidate with a job to start their application journey.
JOB REQUISITION
Select Job
CANDIDATE
Select Candidate
Add to Pipeline
0
dates
INTERVIEW
0
No candidates
OFFER
0
No candidates
HIRED
1
Abhiram Ravella
Full Stack Developer
SCORE: 89
View
REJECTED
Santhosh K V
SCORE: 98


Select one or more roles. Roles control what pages and actions this user can access.
Admin
Applicant
Hiring Manager
Interviewer
Recruiter
Create User
Team Members
7 USERS
NAME
EMAIL
ROLES
ACTIVE
Test Applicant
privatesanthoshvedakrishnan@gmail.com
Applicant
Abhiram Ravella
abhiramravella@gmail.com
Applicant
Santhosh K V
santhosh_k@srmap.edu.in
Applicant
Kurra Swetha
lakshmiswetha_kurra@srmap.edu.in
Interviewer
Mohith Virigineni
mohithv752006@gmail.com
Recruiter
Santhosh
santhoshvedakrishnan@gmail.com
Hiring Manager
Platform Admin
admin@hirematrix.local
Admin


Candidate Profiles
3 CANDIDATES
Test Applicant
Edit
Delete
privatesanthoshvedakrishnan@gmail.com
9361888416
0 yrs exp
React
Node.js
Express
MongoDB
REPLACE RESUME
Choose PDF or Word doc
Upload & Score
Resume on file
Abhiram Ravella
Edit
Delete
abhiramravella@gmail.com
7418529630
0 yrs exp
react.js
MySql
REPLACE RESUME
Choose PDF or Word doc
Upload & Score
Resume on file
Santhosh K V
Edit
Delete
santhosh_k@srmap.edu.in
0 yrs exp
REPLACE RESUME
Choose PDF or Word doc
Upload & Score
Resume on file


HireMatrix Applicant Portal
Track your applications and interviews
Logout
Dashboard
Jobs
Notifications
My Applications
Track your current stage and interview schedule in one place.
Profile
Test Applicant
privatesanthoshvedakrishnan@gmail.com
9361888416
Experience: 0 years
UPLOAD LATEST RESUME
Choose Resume
No file selected
Upload Resume
Last uploaded: 02/05/2026, 03:23 pm
Upcoming Interviews
No upcoming interviews yet.


Gmail
in:sent
Compose
Inbox 3,433
Starred
Snoozed
Sent
Drafts 9
Purchases 40
More
Labels
1 of 350
Stage updated: Abhiram Ravella → Screening Inbox ×
santhoshvedakrishnan@gmail.com
to me
3:26 PM (11 minutes ago)
An application stage has been updated.
Candidate: Abhiram Ravella
Role: Full Stack Developer (Engineering)
Stage Change: Applied → Screening
Updated At: 02/05/2026, 03:26 pm
Application ID: 69e5c9f1f399d759deb95477
Reply
Forward

15. Conclusion

HireMatrix represents a comprehensive solution for modern recruitment challenges, combining robust workflow management with intelligent automation. The system successfully addresses the core needs of hiring teams: centralized tracking, role-based collaboration, candidate experience, and data-driven decision making.

The architecture follows industry best practices with clear separation of concerns, scalable design patterns, and modern technology choices. The use of React 19 with Vite provides a fast, responsive frontend experience, while the Node.js/Express backend with MongoDB offers flexibility and performance for handling recruitment data.

Key technical achievements include the AI-powered resume scoring system that leverages Google Generative AI for intelligent candidate evaluation, the queue-based asynchronous processing architecture that ensures system responsiveness, and the comprehensive role-based access control that enables secure multi-stakeholder collaboration.

The platform's strength lies in its end-to-end coverage of the hiring lifecycle, from job creation through final decision, with features that automate repetitive tasks, standardize evaluation processes, and provide visibility to all stakeholders. The integration with Google Calendar and Meet streamlines interview coordination, while the notification system keeps all parties informed throughout the process.

Future enhancements could include advanced analytics and reporting, integration with additional calendar providers, mobile application development, and machine learning improvements for more accurate candidate-job matching. The current architecture provides a solid foundation for such extensions.

For organizations seeking to modernize their hiring process, reduce time-to-hire, and improve the quality of hiring decisions, HireMatrix offers a powerful, scalable, and maintainable solution built on proven technologies and best practices.